

Assesment Tool-1

95%

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Course:- Design and Analysis of Algorithms.

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Binary Search - Middle element analysis

Problem interpretation

Given sorted dataset:

[6, 12, 18, 24, 30, 36, 42]

Find the element 24 using the Binary Search algorithm.

Why binary search is suitable

Binary Search works only on sorted data. Since the given dataset is arranged in ascending order, Binary Search can repeatedly divide the search interval into two halves, making it much faster than Linear search.

Step-by-step Binary Search

step	search interval	mid index	Mid element	comparison	result
1	0-6	$(0+6)/2=3$	24	$24=24$	found

No. of Comparisons

Comparison 1: $24 == 24$

Total = 1

Time complexity

Best case: $O(1)$. Target found at middle in the first comparison).

• Avg Case : $O(\log_2 n) \approx 2.81 \approx 3$

• Worst case : $O(\log_2 n) \approx 2.81 \approx 3$.

For $n=7$, the max comparisons are about $\log_2 7 \approx 3$.

Verification of Correctness.

middle element of dataset = 24.

Matches target value.

∴ Binary search correctly finds the element at the index 3. (or 4th position).

Efficiency Interpretation.

Since the target element already middle of the sorted dataset, Binary Search finds it in only one comparison, which is the best-case scenario. This demonstrates the high efficiency of Binary Search, as it avoids unnecessary comparisons and quickly locates the required element.

• Dataset Size = 7

• Elements Checked = 1.

• Elements not checked = 6.

• Search space reduced = 100%. in the first comparison (target found immediately).

• Efficiency = Best-case Binary Search $O(1)$.